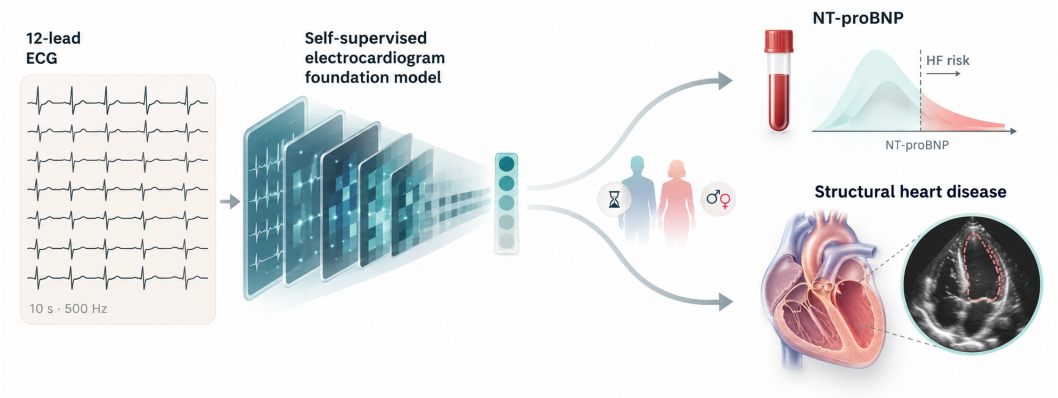
HARVI-ECG: Heart-Failure Readouts from a 10-Second 12-Lead ECG

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Abstract

HARVI-ECG turns a standard 10-second 12-lead ECG into heart-failure readouts. It is built on an ECG foundation encoder pretrained without labels on 6,009,279 ECGs from 3,257,195 patients, and provides two readouts: an *NT-proBNP readout* (an NT-proBNP estimate with rule-out, age-adjusted rule-in and severe decisions) and a *structural heart disease (SHD) module* that detects echo-defined structural heart disease. On MIMIC-IV (18,036 ECG–NT-proBNP pairs, 11,929 patients, patient-grouped 5-fold cross-validation), the NT-proBNP readout reached a mean rule-out/rule-in AUROC of 0.894 (0.889–0.898) and a rule-out AUROC of 0.920 (0.914–0.925). Without retraining it transferred to two hospitals it had never seen: at Emory it detected HFrEF with AUROC 0.810 (0.803–0.816) and HFpEF with 0.727 (0.719–0.735) against age- and sex-matched controls, and at EchoNext it identified echocardiographic reduced ejection fraction with AUROC 0.834. Foundation pretraining outperformed a task-specific model trained from scratch, most clearly at the unseen hospital (Emory HFrEF +0.036 (+0.031 to +0.040), HFpEF +0.055 (+0.049 to +0.060)). Adapted on EchoNext echocardiography labels, the SHD module reached AUROC 0.929 for reduced ejection fraction and 0.850 for composite structural heart disease on held-out patients, and needed only 250 labelled patients to match the zero-label NT-proBNP readout for LVEF $\leq$45 %.



Overview. A 10-second 12-lead ECG is encoded by a self-supervised ECG foundation model; HARVI-ECG reads out NT-proBNP-based heart-failure risk and echo-defined structural heart disease.

Technical report v1.2, October 2026. Code: https://github.com/CellularSyntax/harvi-ecg-eval. Archive: doi:10.5281/zenodo.23367346.

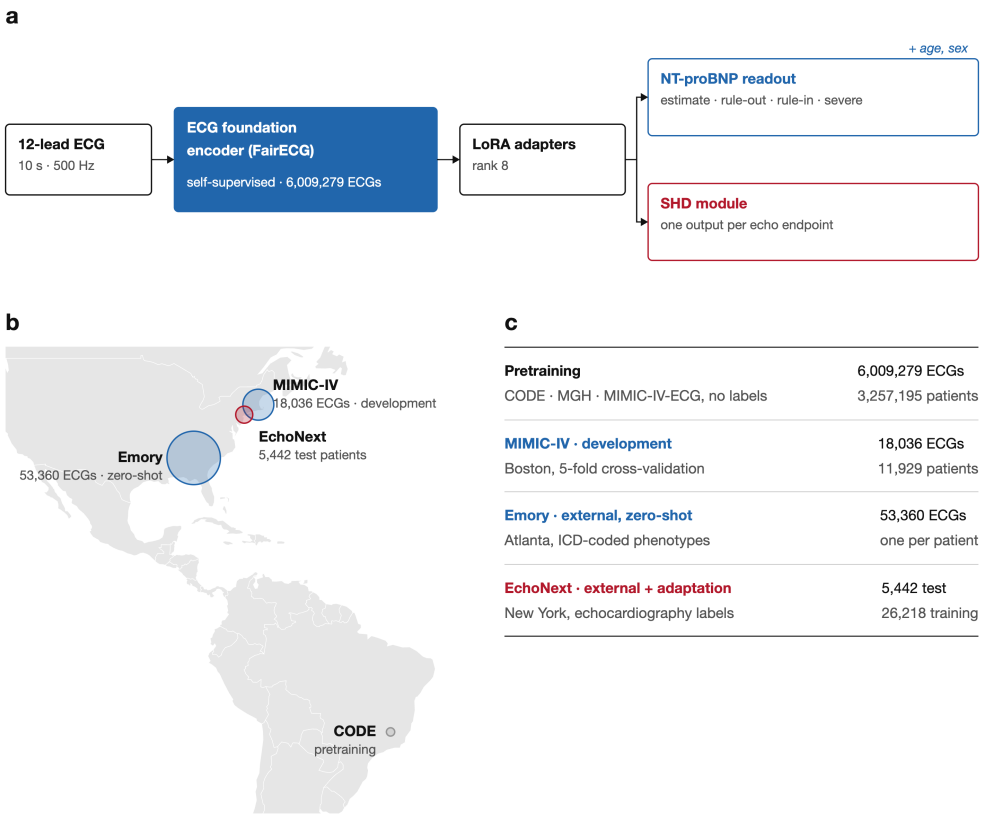


Figure 1: HARVI-ECG and its cohorts. (a) Architecture; age and sex feed only the NT-proBNP readout. (b) Cohort sites; circle area is proportional to cohort size. The Boston circle marks MIMIC-IV; the pretraining sources MGH and MIMIC-IV-ECG are also Boston-based and are listed in (c). (c) Cohort roles and counts. Emory and EchoNext were used neither in pretraining nor in developing the NT-proBNP readout.

1 Model

HARVI-ECG (Figure 1a) has three parts.

- **ECG foundation encoder.** The FairECG encoder (internal, ongoing unpublished work), an xLSTM-based model [4] pretrained with self-supervision (SimDINO) on 6,009,279 ECGs from 3,257,195 patients (CODE, MGH, MIMIC-IV-ECG; waveforms only, no labels). It maps a 10-second, 500-Hz, 12-lead ECG (each lead instance-normalised) to a 1,024-dimensional embedding.
- **NT-proBNP readout.** The encoder is adapted with low-rank adapters (LoRA, rank 8) [5] and feeds a shared head that also receives age and sex. The head outputs an NT-proBNP estimate with uncertainty and three guideline decisions: rule-out (<300 pg/mL), age-adjusted rule-in (>450/900/1,800 pg/mL for <50/50–75/>75 years) and severe (>3,000 pg/mL). It is trained on MIMIC-IV only; the epoch is selected on an inner 10 % of training patients.
- **SHD module.** The same encoder with LoRA and one logistic output per echocardiographic endpoint, adapted on the echo-labelled patients of a site (here EchoNext).

A task-specific baseline (ConvNeXt-1D + xLSTM with age and sex conditioning, trained

Table 1: MIMIC-IV out-of-fold results, all patients (n = 18,036 ECGs, 11,929 patients). AUROC with 95 % patient-bootstrap CI (2,000 draws). Δ is the paired difference to the task-specific baseline on the pre-specified decision metric.

Model	Rule-out <300	Rule-in (age-adj.)	Severe >3,000	Mean rule-out + rule-in	Δ mean vs baseline
Task-specific baseline	0.902 0.896–0.908	0.863 0.856–0.869	0.863 0.856–0.869	0.882 0.877–0.888	—
HARVI-ECG	0.920 0.914–0.925	0.868 0.862–0.874	0.848 0.841–0.855	0.894 0.889–0.898	+0.011 +0.007 to +0.015

from scratch on the same MIMIC-IV folds) quantifies the benefit of foundation pretraining (Section 4). Analyses followed pre-specifications fixed before scoring; the primary decision metric was the mean of the rule-out and rule-in AUROCs, with 95 % CIs from a 2,000-draw patient bootstrap.

2 Data

MIMIC-IV (development). 18,036 ECG–NT-proBNP pairs from 11,929 patients of the MIMIC-IV-ECG matched subset [1, 2]. Median age 72 years (IQR 60–82); 50.4 % female; NT-proBNP median 2,046 pg/mL (IQR 515–6,276). Prevalence: rule-out 18.1 %, rule-in 61.6 %, severe 41.1 %. Folds are `GroupKFold(5)` on patient.

Emory (external). Site I0006 of the Harvard-Emory ECG Database [3], used neither for development nor for pretraining. HFrEF (ICD-10 I50.2x) and HFpEF (I50.3x) cases, with the ECG closest to the first subtype code (−180 to +30 days), against controls without any I50 code and at least 365 days of follow-up, matched 4:1 on sex and 5-year age band. HFrEF: 5,627 cases, 22,508 controls. HFpEF: 5,045 cases, 20,180 controls. One ECG per patient (53,360 in total). The stored aVL and aVF channels were rebuilt from leads I and II before scoring.

EchoNext (external). 12-lead ECGs paired with transthoracic echocardiography (Columbia University), with patient-disjoint training, validation and test splits; not used in pretraining. All EchoNext results refer to the 5,442 test patients (one ECG each). Endpoints: composite structural heart disease (moderate or greater, EchoNext definition), LVEF ≤45 %, LV wall thickness ≥13 mm and reduced ejection fraction (LVEF ≤40 % vs ≥50 %; post hoc).

3 NT-proBNP readout

Development. On MIMIC-IV, HARVI-ECG reached AUROC 0.920 (0.914–0.925) for rule-out, 0.868 (0.862–0.874) for rule-in and 0.848 (0.841–0.855) for the severe decision; the mean of rule-out and rule-in was 0.894 (0.889–0.898) (Table 1, Figure 2a–c). The estimate ranks patients consistently with measured NT-proBNP (Spearman $\rho = 0.731$; Figure 2g). At the default operating point, the rule-out decision has sensitivity 0.780 and specificity 0.885 (Figure 2h). 9,506 of the 11,929 patients had ECG waveforms in the pretraining corpus; on the 2,423 patients without, the mean rule-out/rule-in AUROC was 0.897 (0.887–0.907) (Table 2).

Unseen hospitals. Applied unchanged, the NT-proBNP estimate detected HFrEF at Emory with AUROC 0.810 (0.803–0.816) and HFpEF with 0.727 (0.719–0.735) (Table 3, Figure 2d,e). At EchoNext it identified reduced ejection fraction with AUROC 0.834 and LVEF ≤45 % with 0.830 (Figure 2f). How the fixed pg/mL cut-offs transfer is analysed in Section 6.

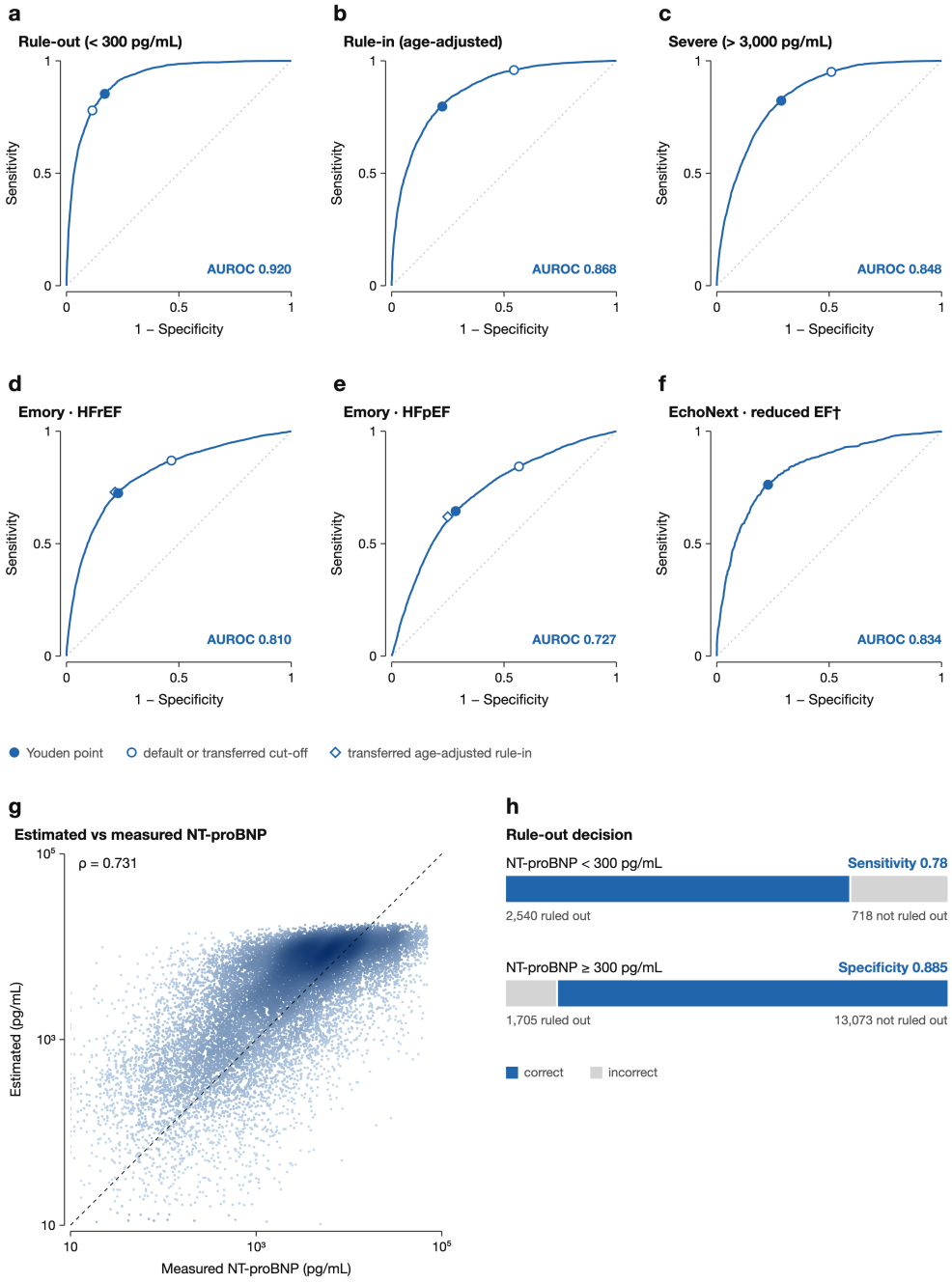


Figure 2: NT-proBNP readout. (a–c) MIMIC-IV out-of-fold ROC curves for rule-out (<300 pg/mL), age-adjusted rule-in and severe (>3,000 pg/mL). (d, e) Emory, zero-shot: ICD-coded HFrEF and HFpEF against age- and sex-matched controls. (f) EchoNext held-out test patients, reduced EF†, zero-shot. Filled circles: Youden points with thresholds chosen on separate data (Section 6); open circles: default (head probability ≥0.5) or transferred 300 pg/mL cut-off; open diamonds: transferred age-adjusted rule-in cut-offs. (g) Estimated versus measured NT-proBNP per ECG, coloured by density (log axes; ECGs with estimates at the 10 pg/mL floor not shown; dashed: line of identity). (h) Rule-out decision at the default operating point; positive class = measured NT-proBNP <300 pg/mL. †LVEF ≤40 % vs ≥50 %, post hoc.

Table 2: MIMIC-IV out-of-fold results, patients whose ECGs were not in the pretraining corpus (n = 3,674 ECGs, 2,423 patients). AUROC with 95 % patient-bootstrap CI (2,000 draws). Δ is the paired difference to the task-specific baseline on the pre-specified decision metric.

Model	Rule-out <300	Rule-in (age-adj.)	Severe >3,000	Mean rule-out + rule-in	Δ mean vs baseline
Task-specific baseline	0.900 0.887–0.913	0.864 0.849–0.878	0.867 0.853–0.880	0.882 0.871–0.893	—
HARVI-ECG	0.924 0.912–0.934	0.871 0.857–0.884	0.851 0.836–0.866	0.897 0.887–0.907	+0.015 +0.007 to +0.024

Table 3: Zero-shot results at Emory (score: NT-proBNP estimate, mean of the five fold models). Sensitivity and specificity at the fixed MIMIC-IV cut-offs (300 pg/mL and age-adjusted rule-in).

Endpoint	Model	AUROC (95 % CI)	Δ vs baseline	Sens / spec at 300	Sens / spec at rule-in
HFrEF vs controls	Baseline	0.774 0.767–0.781	—	0.98 / 0.11	0.89 / 0.47
	HARVI-ECG	0.810 0.803–0.816	+0.036 +0.031 to +0.040	0.87 / 0.53	0.73 / 0.78
HFpEF vs controls	Baseline	0.672 0.664–0.680	—	0.98 / 0.07	0.81 / 0.43
	HARVI-ECG	0.727 0.719–0.735	+0.055 +0.049 to +0.060	0.84 / 0.43	0.62 / 0.75

4 Foundation pretraining versus a task-specific model

Under the pre-specified rule, HARVI-ECG improved the mean rule-out/rule-in AUROC over the task-specific baseline by +0.011 (+0.007 to +0.015), with the CI excluding zero (Figure 3, Table 1). The gain was largest at the unseen hospital: Emory HFrEF +0.036 (+0.031 to +0.040) and HFpEF +0.055 (+0.049 to +0.060). The baseline could not be applied at EchoNext, whose waveforms are released per-lead standardised; HARVI-ECG normalises each lead itself and is unaffected.

5 SHD module

Adapted on EchoNext echocardiography labels (26,218 training patients; mean of three seeds), the SHD module outperformed the zero-shot NT-proBNP readout on every endpoint (Table 4, Figure 4a,b). It reached AUROC 0.929 for reduced ejection fraction, 0.918 for LVEF ≤45 % (0.943 in outpatients) and 0.850 for composite structural heart disease. The largest gain was for LV wall thickness (+0.100 (+0.085 to +0.115)), the phenotype NT-proBNP reflects least.

Few labels are needed (Figure 4c,d). A lightweight adaptation of the encoder matched the zero-label NT-proBNP readout with 250 labelled patients for LVEF ≤45 % and LV wall thickness and with 1,000 for structural heart disease, and kept improving with more labels (LVEF ≤45 %: 0.911 with all training patients).

The two readouts are complementary: the NT-proBNP readout works at a new site without local labels, and a few hundred echo-labelled patients enable a dedicated structural heart disease screener.

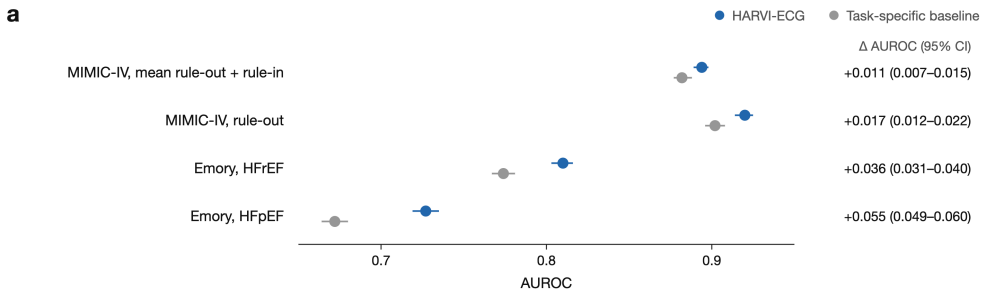


Figure 3: HARVI-ECG against the task-specific baseline (trained from scratch on the same MIMIC-IV folds), on identical patients. AUROC with 95 % patient-bootstrap CI; right column: paired ΔAUROC (95 % CI).

Table 4: EchoNext held-out test patients. AUROC with 95 % patient-bootstrap CI and paired difference (SHD module − NT-proBNP readout). [a]Composite of moderate-or-greater echo findings. [b]Post hoc endpoint.

Endpoint	Cases / n	NT-proBNP readout	SHD module	Δ
Structural heart disease[a]	2,318 / 5,442	0.808 0.796–0.820	**0.850** 0.840–0.860	+0.042 +0.034 to +0.049
LVEF ≤45 %	962 / 5,442	0.830 0.816–0.844	**0.918** 0.907–0.928	+0.088 +0.076 to +0.099
LVEF ≤40 vs ≥50 %[b]	761 / 4,492	0.834 0.819–0.849	**0.929** 0.917–0.940	+0.095 +0.082 to +0.108
LV wall ≥13 mm	1,061 / 5,442	0.685 0.667–0.702	**0.785** 0.771–0.799	+0.100 +0.085 to +0.115

6 Operating points

Each threshold was chosen on data separate from the patients on which it is evaluated: in MIMIC-IV by cross-fitting (the Youden threshold of the other four folds, applied to the held-out fold), at Emory by splitting the patients into halves (Youden threshold from one half applied to the other; median of 20 repeats), and at EchoNext on the validation split. Confidence intervals come from 2,000 patient-bootstrap draws with the threshold held fixed.

In MIMIC-IV, the Youden thresholds give balanced sensitivity and specificity for all three decisions (rule-out 0.85/0.83, rule-in 0.80/0.77, severe 0.82/0.71; Figure 5a), whereas the default threshold favours sensitivity for rule-in and severe. At Emory, the age-adjusted rule-in cut-offs transferred from MIMIC-IV performed as well as a threshold recalibrated at Emory itself (HFrEF 0.73/0.78 versus 0.72/0.77; Figure 5b). The fixed 300 pg/mL rule-out cut-off flagged most patients at Emory (specificity 0.53 for HFrEF and 0.43 for HFpEF) and is the cut-off that needs local recalibration. On EchoNext, at validation-chosen thresholds, the SHD module reached sensitivity/specificity 0.84/0.86 for reduced ejection fraction, against 0.76/0.77 for the NT-proBNP readout (Figure 5c).

7 Comparison with published models

HARVI-ECG is on par with published ECG-to-NT-proBNP models (Figure 6) and additionally provides zero-shot validation at two unseen hospitals and a structural heart disease readout.

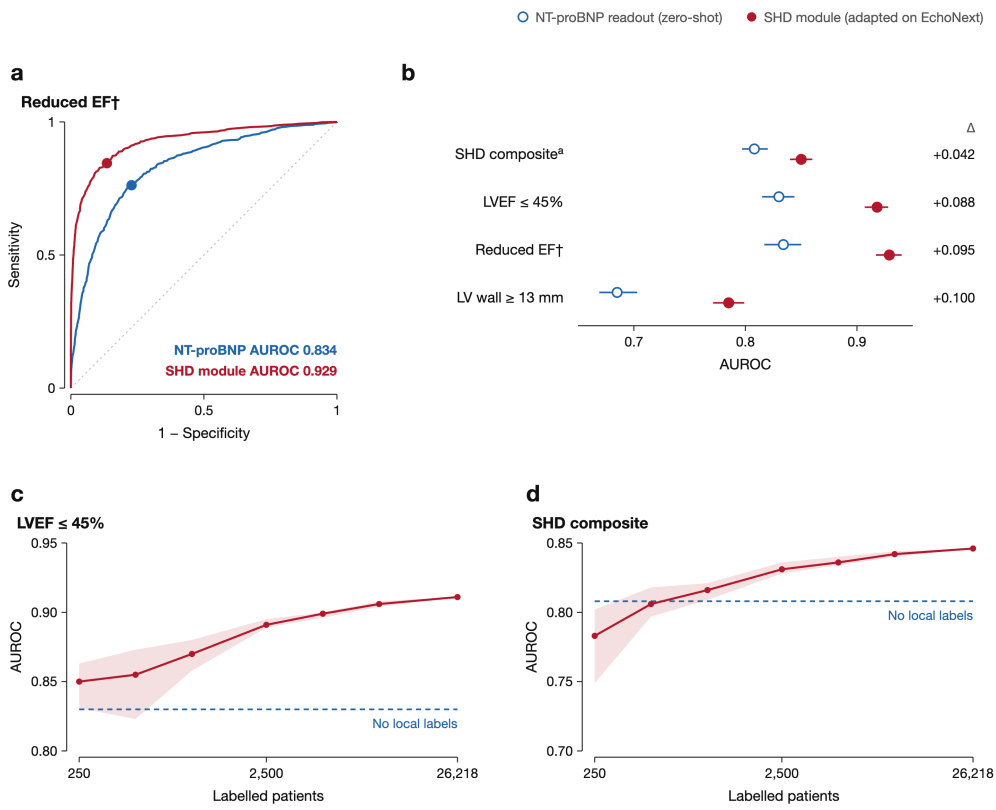


Figure 4: SHD module on held-out EchoNext test patients. (a) ROC for reduced EF†: NT-proBNP readout (zero-shot) and SHD module (adapted on EchoNext); filled circles are Youden points chosen on the validation split. (b) AUROC (95 % CI) per endpoint; right column: paired Δ. (c, d) AUROC of a lightweight adaptation (linear probe) against the number of echo-labelled patients (log scale); line: mean, band: min–max of 5 draws; dashed: NT-proBNP readout with no local labels. [a]Moderate or greater echo finding. †Post hoc.

8 Limitations

1. Retrospective evaluation; prospective validation of the screening pathway is planned.
2. Emory phenotypes are ICD-10 based. EchoNext records echo findings but not symptoms, so its positives include symptomatic heart failure as well as asymptomatic structural disease (Stage B).
3. The 300 pg/mL rule-out cut-off requires site recalibration (Section 6). The NT-proBNP regression head was trained on the raw pg/mL scale (the pre-specification described a log scale); a log-scale head is planned. The decision rule uses the classification outputs and is unaffected.
4. Each fold model of the NT-proBNP readout was trained with a single seed.

9 Availability

Analysis code, aggregate results and figures are available in the accompanying repository and Zenodo record. Model weights are available upon reasonable request through a gated Hugging Face repository (research use only; not a medical device). MIMIC-IV and HEEDB

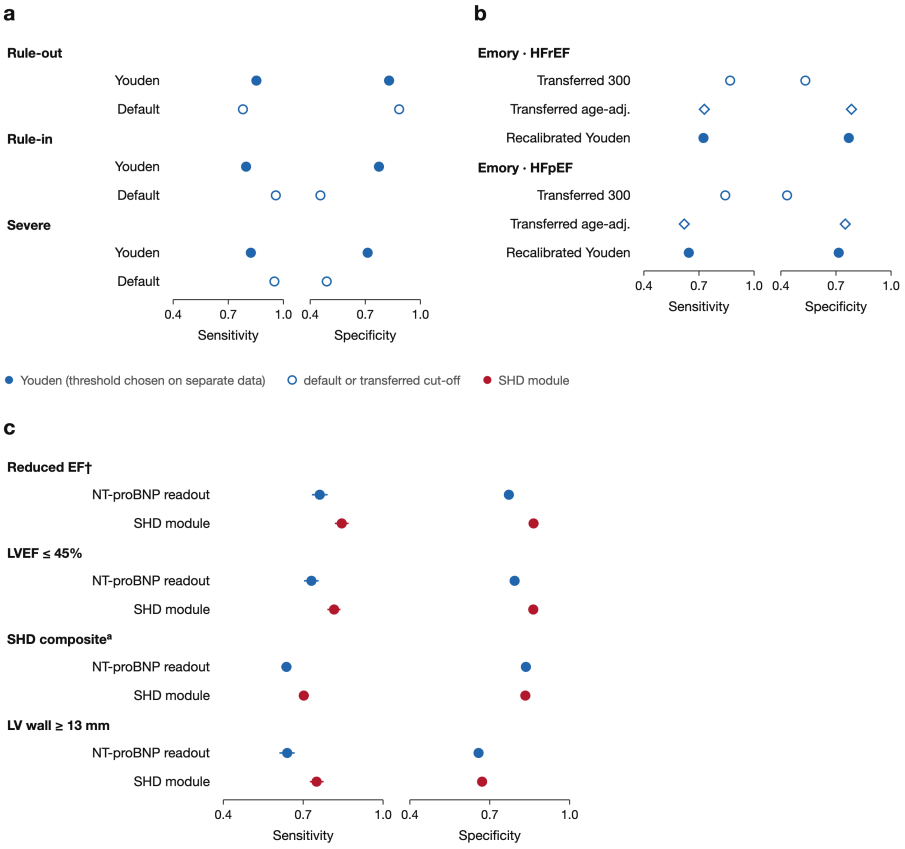


Figure 5: Sensitivity and specificity at operating points (95 % CI, patient bootstrap, threshold fixed). (a) MIMIC-IV: cross-fitted Youden threshold (filled) and default threshold (open). (b) Emory: MIMIC-IV cut-offs transferred unchanged (open circle: 300 pg/mL; open diamond: age-adjusted rule-in) and a Youden threshold recalibrated at Emory (filled). (c) EchoNext test: Youden thresholds chosen on the validation split, for both readouts. Values are listed in Table 5 (Appendix A). [a]Moderate or greater echo finding. †Post hoc.

are credentialed-access datasets and EchoNext is available under its data-use agreement; patient-level data and predictions are not redistributed.

References

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[2] A. E. W. Johnson, L. Bulgarelli, L. Shen, et al. MIMIC-IV, a freely accessible electronic health record dataset. *Scientific Data*, 10:1, 2023.

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[4] M. Beck, K. Pöppel, M. Spanring, et al. xLSTM: Extended long short-term memory. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. arXiv:2405.04517.

[5] E. J. Hu, Y. Shen, P. Wallis, et al. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations (ICLR)*, 2022. arXiv:2106.09685.

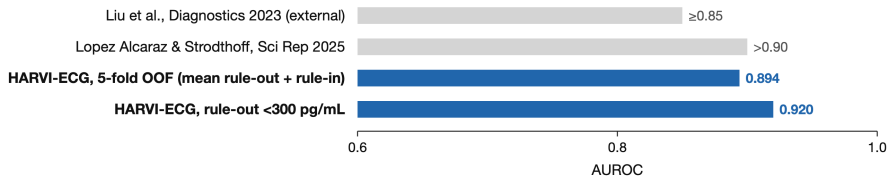


Figure 6: HARVI-ECG against published ECG-to-NT-proBNP models. Published values as reported by the cited studies, on different cohorts; the axis starts at 0.6.

A Operating points

Table 5: Operating points (values shown in Figure 5). Sensitivity and specificity with 95 % CI (2,000 patient-bootstrap draws, threshold fixed). MIMIC-IV probability thresholds are the median of the five fold thresholds. For rule-out, the positive class is measured NT-proBNP <300 pg/mL. [a]Moderate or greater echo finding. [†]LVEF ≤40 % vs ≥50 %, post hoc.

Operating point	Threshold	Sensitivity	Specificity
MIMIC-IV, out-of-fold (NT-proBNP readout)			
Rule-out, Youden (cross-fitted)	$p \geq 0.33$	0.85 (0.84–0.87)	0.83 (0.82–0.84)
Rule-out, default	$p \geq 0.50$	0.78 (0.76–0.80)	0.88 (0.88–0.89)
Rule-in, Youden (cross-fitted)	$p \geq 0.89$	0.80 (0.79–0.81)	0.77 (0.76–0.79)
Rule-in, default	$p \geq 0.50$	0.96 (0.95–0.96)	0.46 (0.44–0.47)
Severe, Youden (cross-fitted)	$p \geq 0.84$	0.82 (0.81–0.83)	0.71 (0.70–0.72)
Severe, default	$p \geq 0.50$	0.95 (0.95–0.96)	0.49 (0.48–0.50)
Emory, zero-shot (NT-proBNP readout)			
HFrEF, transferred 300 pg/mL	300 pg/mL	0.87 (0.86–0.88)	0.53 (0.53–0.54)
HFrEF, transferred age-adjusted rule-in	450/900/1,800 pg/mL	0.73 (0.72–0.74)	0.78 (0.78–0.79)
HFrEF, recalibrated Youden	992 pg/mL	0.72 (0.71–0.74)	0.77 (0.76–0.78)
HFpEF, transferred 300 pg/mL	300 pg/mL	0.84 (0.83–0.85)	0.43 (0.43–0.44)
HFpEF, transferred age-adjusted rule-in	450/900/1,800 pg/mL	0.62 (0.61–0.63)	0.75 (0.74–0.76)
HFpEF, recalibrated Youden	1,057 pg/mL	0.64 (0.63–0.66)	0.71 (0.71–0.72)
EchoNext, held-out test (Youden threshold from the validation split)			
Reduced EF[†], NT-proBNP readout	2,203 pg/mL	0.76 (0.73–0.79)	0.77 (0.76–0.79)
Reduced EF[†], SHD module	$p \geq 0.25$	0.84 (0.82–0.87)	0.86 (0.85–0.88)
LVEF ≤45 %, NT-proBNP readout	2,203 pg/mL	0.73 (0.70–0.76)	0.79 (0.78–0.80)
LVEF ≤45 %, SHD module	$p \geq 0.24$	0.82 (0.79–0.84)	0.86 (0.85–0.87)
SHD composite[a], NT-proBNP readout	1,406 pg/mL	0.64 (0.62–0.66)	0.84 (0.82–0.85)
SHD composite[a], SHD module	$p \geq 0.53$	0.70 (0.68–0.72)	0.83 (0.82–0.85)
LV wall ≥13 mm, NT-proBNP readout	1,155 pg/mL	0.64 (0.61–0.67)	0.66 (0.64–0.67)
LV wall ≥13 mm, SHD module	$p \geq 0.20$	0.75 (0.73–0.78)	0.67 (0.66–0.68)